Extending Theorem 2 of *Robust Trust* to infinite M and Θ:
A three-tier theorem via the payoff-profile menu engine

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Abstract

This note reports on a proof, drafted over the past week, of a three-tier infinite-dimensional extension of Theorem 2 ("Robustly Rationalizable Solution") from Dworczak & Smolin's *Robust Trust* (2026). The earlier Phil-Reny restricted-game-plus-Lusin-lift route, which required three additional hypotheses (**A5-thick**, **A8c-attain**, **TRE-gen-Hall**) even for value optimality, has been replaced by a finite-dimensional *payoff-profile menu engine*. The reformulation is strictly stronger in three respects. (i) **Tier 1a** (existence of a value-optimal agent strategy plus ε-adversaries) holds under the paper's standing hypotheses alone; A5-thick and A8c-attain are no longer needed. (ii) **Tier 1b** (exact adversary attainment) is now isolated behind a clean *exact-contact* condition endogenous to the chosen optimal labeling. (iii) **Tier 2** (full robust rationalizability) replaces deterministic TRE-gen-Hall with a set-valued *menu-Hall* calibration. A ternary non-radial witness shows menu-Hall is genuinely needed in $|\Omega| \geq 3$: the obstruction is intrinsic multi-dimensional vector balance, not a deterministic-versus-randomized artefact.

1 The setting and the question

We use the notation of *Robust Trust*. Ω is finite with full-support prior μ_0; $s \in \Delta(\Omega)$ has state-conditional law $\pi(\cdot \mid \omega)$ and unconditional law $\tau = \sum_\omega \mu_0(\omega)\pi(\cdot \mid \omega)$; $M = \text{supp}(\tau)$; $\theta \in \Theta$ (compact metric) is drawn from $f(\cdot \mid \omega)$, independently of s conditional on ω; A is compact metric; and $u(a, \omega, \theta)$ is bounded and continuous in a. Let Σ denote the agent's measurable strategies $\sigma : M \times \Theta \to \Delta(A)$ and B the misaligned adviser's measurable kernels $\beta : M \to \Delta(M)$. With probability α the adviser is aligned and reports s truthfully; with probability $1-\alpha$ the adviser is misaligned.

The agent's robust objective is

$$U(\sigma) \;=\; \alpha\,\mathbb{E}_{\text{id},\sigma}[u] + (1-\alpha)\,\inf_{\beta \in B} \mathbb{E}_{\beta,\sigma}[u], \qquad U^* \;=\; \sup_{\sigma \in \Sigma} U(\sigma), \tag{1}$$

and Definition 2 (*robust rationalizability*) asks for (σ^*, β^*) such that, at every on-path message m, the agent's prescribed continuation is Bayes-optimal under $P_{\beta^*}(\cdot \mid m)$. The paper's Theorem 2 establishes existence under the assumption that M and Θ are *finite*, via a finite Sion saddle. The question addressed here is whether existence can be extended when M and Θ are infinite.

2 The engine

The earlier route worked in the agent's strategy space with the Balder weak topology on (Σ, T_λ), used Mertens's asymmetric minmax theorem on the restricted game, and invoked Lusin regularization to lift the restricted maximizer to the unrestricted game. That machinery is heavy and required three added hypotheses, the most consequential being A5-thick (a Lusin-thickness clause imposed on a representative of the maximizer).

The current route works instead in the finite-dimensional *payoff-profile space*

$$W \ \subseteq\ \mathbb{R}^{|\Omega|}, \qquad W \ :=\ \{w \in \mathbb{R}^{|\Omega|} : w(\omega) = \mathbb{E}_{\hat\sigma}[u(a,\omega,\theta) \mid \omega] \text{ for some } \hat\sigma \in \hat\Sigma\},$$

where $\hat\Sigma$ is the space of private strategies $\hat\sigma : \Theta \to \Delta(A)$. W is a compact convex subset of $\mathbb{R}^{|\Omega|}$. The agent's optimization reduces to choosing a Borel labeling $w : M \to W$ (a payoff-profile strategy); the misaligned adviser optimizes over Borel kernels $\beta : M \to \Delta(M)$; the value of the game depends on w only through the compact *menu* $C := \overline{w(M)} \subseteq W$.

This finite-dimensional reformulation has three advantages. First, the space $\mathcal{K}(W)$ of nonempty compact subsets of W is itself compact under the Hausdorff metric, so existence of an optimal menu C^* comes for free without any thickness or representative-choice hypothesis. Second, a closure-pruning step cleanly identifies the support $C^\dagger := \overline{w^*(M)} \subseteq C^*$ of an optimal labeling, decoupling existence from contact-set geometry. Third, the rowwise contact set

$$G(s) \ :=\ \{m \in M : s \cdot w^*(m) = \min_{z \in C^\dagger} s \cdot z\}$$

becomes a finite-dimensional object whose existence and selectability properties are easy to state and easy to fail in tight examples.

3 Main theorem

The standing hypotheses of *Robust Trust* are kept verbatim (Ω finite, μ_0 full support, A, Θ compact metric, u bounded continuous in a, conditional independence of s, θ given ω, all measurability Borel). The new theorem has three tiers, with the second and third each layering one additional endogenous condition.

Tier 1a (standing hypotheses alone). There exists $\sigma^* \in \Sigma$ with $U(\sigma^*) = U^*$. For every $\varepsilon > 0$, there exists $\beta_\varepsilon \in B$ with $U(\beta_\varepsilon, \sigma^*) \le U^* + \varepsilon$.

Exact-contact condition. With w^* the chosen optimal labeling and $C^\dagger := \overline{w^*(M)}$, the rowwise contact set $G(s)$ is nonempty for τ-a.e. s and $s \mapsto G(s)$ admits a Borel measurable selector $m^* : M \to M$ with $m^*(s) \in G(s)$ τ-a.e.

Tier 1b (+ exact-contact). With $\beta^*(dm \mid s) := \delta_{m^*(s)}(dm)$, $U(\beta^*, \sigma^*) = \inf_{\beta \in B} U(\beta, \sigma^*) = U^*$.

Menu-Hall condition. There exists a set-valued kernel $\kappa : M \to \Delta(M)$ supported on $G(s)$ (i.e. $\kappa(G(s) \mid s) = 1$ for τ-a.e. s) such that, for the joint coupling $\gamma_\alpha := \alpha\,(\mathrm{id}, \mathrm{id})_{\#}\tau + (1-\alpha)\,\tau \otimes \kappa$ on $M \times M$ with second-marginal $q := (\gamma_\alpha)_2$, the posterior satisfies $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ for q-a.e. m, where $B(m) := \{\mu \in \Delta(\Omega) : \hat\sigma^*(m) \in \arg\max_{\hat\sigma'} U(\hat\sigma', \mu)\}$ is the Bayes cone for the agent's continuation at message m.

Tier 2 (+ menu-Hall). When $\alpha > 0$, $\hat\sigma^*(m) \in \arg\max_{\hat\sigma'} U(\hat\sigma', P_{\gamma_\alpha}(\cdot \mid m))$ for τ-a.e. m. Hence σ^* is robustly rationalizable in the paper's a.e./on-path sense.

4 Proof strategy in one paragraph

Tier 1a. The value of the game depends on w only through the menu C, and the menu-value functional $F(C) := \int_M [\alpha \max_{w \in C} m \cdot w + (1-\alpha) \min_{w \in C} s \cdot w]\, \tau(ds)$ is upper semicontinuous on the Hausdorff-compact $\mathcal{K}(W)$. Hence an optimal C^* exists, an optimal labeling $w^* : M \to C^*$ realizing $\max_{w \in C^*} m \cdot w$ pointwise exists by Aliprantis–Border 18.19, and a profile-realization sub-lemma (KRN selection on $\hat{\Sigma} \twoheadrightarrow W$) lifts w^* to a private strategy $\hat{\sigma}^*$. The closure-pruning step $C^\dagger := \overline{w^*(M)}$ preserves the value because $\min_{w \in C^\dagger} s \cdot w \geq \min_{w \in C^*} s \cdot w$ pointwise and C^* was already optimal. The ε-adversary is built by Jankov–von Neumann selection on the ε-thick approximate contact set $G_\varepsilon(s) := \{m : s \cdot w^*(m) \leq \min_{C^\dagger} s \cdot w + \varepsilon\}$.

Tier 1b. Under exact-contact, replace G_ε by G and the selector m^* delivers $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$ exactly attaining the rowwise minimum τ-a.e.

Tier 2. Under menu-Hall, the support-function form $\int_E \phi(P_{\gamma_\alpha}(\cdot \mid m)) q(dm) \leq \int_E h_{B(m)}(\phi) q(dm)$ for every measurable E and continuous affine ϕ is equivalent (by countable separation in finite-dim $\Delta(\Omega)$) to pointwise membership $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e., which is exactly the Bayes-optimality requirement of Definition 2. Since $q \geq \alpha\tau$ when $\alpha > 0$, q-a.e. upgrades to τ-a.e.

5 Tier 1a: from menu existence to ε-adversaries

Lemma 1 (L1 — menu-value equivalence)**.** *For any Borel labeling $w : M \to W$ with menu $C := \overline{w(M)}$,*

$$U(\sigma_w) \;=\; \alpha \int_M m \cdot w(m)\, \tau(dm) + (1-\alpha) \int_M \min_{z \in C} s \cdot z\, \tau(ds).$$

The first term equals $\int_M \max_{z \in C} m \cdot z\, \tau(dm)$ at any labeling that selects pointwise maximizers on C.

Sketch. The aligned term integrates the agent's payoff at the truthful posterior m against the chosen profile $w(m)$; the misaligned term integrates the rowwise infimum over messages, which equals the infimum over $w(M)$ of $s \cdot w$, which by continuity equals the minimum over $\overline{w(M)}$. □

Lemma 2 (L2 — existence of an optimal menu)**.** *There exists $C^* \in \mathcal{K}(W)$ maximizing $F(C) := \int [\alpha \max_{w \in C} m \cdot w + (1-\alpha) \min_{w \in C} s \cdot w] \tau(ds)$.*

Sketch. $\mathcal{K}(W)$ is compact under Hausdorff distance because W is compact in $\mathbb{R}^{|\Omega|}$. The aligned term is continuous in C (maximum of a continuous function over a Hausdorff-continuous compact set), and the misaligned term is upper semicontinuous in C (minimum is u.s.c. in C). Hence F is u.s.c., and $\mathcal{K}(W)$-compactness gives a maximizer. □

Lemma 3 (L3 — closure-pruning)**.** *Let $w^* : M \to C^*$ be an optimal Borel labeling and $C^\dagger := \overline{w^*(M)}$. Then $F(C^\dagger) = F(C^*) = U^*$.*

Sketch. $w^*(m) \in C^\dagger \subseteq C^*$, so the aligned term is unchanged pointwise. $C^\dagger \subseteq C^*$ implies $\min_{C^\dagger} s \cdot w \geq \min_{C^*} s \cdot w$, so the misaligned term weakly improves. Optimality of C^* forces equality. □

Lemma 4 (L4 — ε-adversary realization)**.** *For every $\varepsilon > 0$, define $G_\varepsilon(s) := \{m \in M : s \cdot w^*(m) \leq \min_{z \in C^\dagger} s \cdot z + \varepsilon\}$. Then $s \mapsto G_\varepsilon(s)$ has analytic graph and admits a universally measurable selector m^*_ε. The kernel $\beta_\varepsilon(\cdot \mid s) := \delta_{m^*_\varepsilon(s)}$ satisfies $U(\beta_\varepsilon, \sigma^*) \leq U^* + (1-\alpha)\varepsilon$.*

Sketch. $G_\varepsilon(s)$ is the preimage under $m \mapsto s \cdot w^*(m)$ of an interval of width ε shifted by an essentially-bounded measurable function $s \mapsto \min_{C^\dagger} s \cdot w$; its graph is analytic. Jankov–von Neumann [2, Thm. 18.13] provides a universally measurable selector. The payoff bound follows by plugging β_ε into the misaligned term of U. □

Tier 1a follows: existence of σ^* by Lemmas 2 and 3 plus the profile-realization sub-lemma; the ε-adversary by Lemma 4.

6 Tier 1b: exact-contact

The exact-contact condition isolates the gap between ε-attainment and exact attainment. There are several sufficient routes, all standard:

- **Closed image.** If $w^*(M) \subseteq W$ is closed, then $C^\dagger = w^*(M)$, the rowwise minimum on $C^\dagger$ is attained on $w^*(M)$, and any minimizer pulls back to $G(s) \neq \emptyset$.
- **Closed-graph correspondence.** If $\Gamma(s) := \arg\min_{w \in C^\dagger} s \cdot w$ is upper hemicontinuous with closed values, the measurable maximum theorem [2, Thm. 18.19] delivers a Borel measurable selector.
- **u.h.c. Bayes-action.** If the agent's Bayes-action correspondence at message m is u.h.c. with closed compact values, then choosing $\hat{\sigma}^*$ as a continuous selector yields a closed labeling, hence closed image.

Under exact-contact, Tier 1b is immediate: $\beta^*(\cdot \mid s) := \delta_{m^*(s)}$ attains $\min_{w \in C^\dagger} s \cdot w$ pointwise τ-a.e., so $U(\beta^*, \sigma^*) = U^*$.

7 Tier 2: menu-Hall calibration

The menu-Hall condition asks that the rowwise minimization can be implemented by a kernel $\kappa(\cdot \mid s)$ whose induced posterior at the messages is Bayes-rationalizable for the agent's continuation $\hat{\sigma}^*(m)$. Two equivalent forms:

- **Kernel form.** $\kappa(G(s) \mid s) = 1$ τ-a.e., and $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e. where $\gamma_\alpha := \alpha(\mathrm{id}, \mathrm{id})_\# \tau + (1-\alpha)\tau \otimes \kappa$ and $q := (\gamma_\alpha)_2$.
- **Support-function form.** For every measurable $E \subseteq M$ and every continuous affine $\phi : \Delta(\Omega) \to \mathbb{R}$,

$$\alpha \int_E \phi(m)\, \tau(dm) + (1-\alpha) \int_M \int_E \phi(s)\, \kappa(dm \mid s)\, \tau(ds) \ \le\ \int_E h_{B(m)}(\phi)\, q(dm),$$

  where $h_{B(m)}(\phi) := \sup_{\mu \in B(m)} \phi(\mu)$.

The support-function inequality, taken over a countable separating family of affine ϕ and all measurable E, is equivalent to the pointwise membership statement $P_{\gamma_\alpha}(\cdot \mid m) \in B(m)$ q-a.e.; equivalence uses finite-dimensional separation in $\Delta(\Omega)$.

This is exactly the per-message Bayes-optimality requirement of Definition 2. Tier 2 follows.

8 Why menu-Hall is needed (sharp witness)

The condition is genuinely needed in $|\Omega| \geq 3$. Take $\Omega = \{0, 1, 2\}$, $A = \{a_0, a_1, a_2\}$ with $u(a_\omega, \omega) = 1$ and $u(a_i, \omega) = -1$ for $i \neq \omega$, prior $\mu_0 = (1/3, 1/3, 1/3)$, atomless full-support τ on $\Delta(\Omega)$, and a non-radial trust region $T := \{\mu : \mu(0) \leq 0.4\}$ with $\hat{\sigma}^*(m)$ the plurality vertex of $P_T(m)$.

The payoff profiles are the three vertices v_0, v_1, v_2 with $v_\omega(\omega) = 1$ and $v_\omega(\omega') = -1$ for $\omega' \neq \omega$. At the boundary message $t_0 := (0.4, 0.3, 0.3)$, the rowwise minimizer set is

$$R(t_0) \;=\; \operatorname*{arg\,min}_{w \in \{v_0, v_1, v_2\}} \; t_0 \cdot w \;=\; \{v_1, v_2\}$$

(check: $t_0 \cdot v_0 = -0.2$, $t_0 \cdot v_1 = t_0 \cdot v_2 = -0.4$). The Bayes cone for the prescribed action $\hat{\sigma}^*(t_0) = a_0$ is $B(t_0) = \{p : p_0 \geq p_1, p_0 \geq p_2\}$. The source beliefs that can adversarially route mass into the t_0-label satisfy $K_0^- := \{s : s_0 \leq s_1, s_0 \leq s_2\}$.

Barycenter argument. Any conditional barycenter $\bar{s}$ of positive τ-mass from K_0^- inherits the pointwise inequalities $s_1 - s_0 \geq 0$ and $s_2 - s_0 \geq 0$, so $\bar{s}_1 \geq \bar{s}_0$ and $\bar{s}_2 \geq \bar{s}_0$. To calibrate into $B(t_0)$, the reverse inequalities are required: $\bar{s}_0 \geq \bar{s}_1$ and $\bar{s}_0 \geq \bar{s}_2$. Together with $\sum \bar{s}_i = 1$, this forces $\bar{s} = (1/3, 1/3, 1/3)$. Pointwise nonneg differences plus equality of barycenter then force the source belief to equal $(1/3, 1/3, 1/3)$ a.s. on the transported mass set. With atomless τ no such concentration is possible. Hence menu-Hall fails.

The diagnosis is sharp: the obstruction is intrinsic multi-dimensional vector balance, not the deterministic-vs-set-valued dichotomy. Set-valued mixing over $R(t_0) = \{v_1, v_2\}$ does not help, because the source-side calibration depends on $\bar{s}$, not on the menu-side selection. Tier 2 therefore requires either menu-Hall as stated or an additional structural hypothesis on the trust region (radial symmetry, zonotopal alignment, separable cones). The ternary radial case (Section 5.2 plus Appendix A.10 of *Robust Trust*) does verify menu-Hall via per-direction 1-dimensional transports.

9 Comparison to the earlier Phil-Reny route

Conclusion	**Earlier route (Phil-Reny)**	**Current route (menu engine)**
σ^* optimal	standing + A5-thick	standing alone
ε-adversary	standing + A5-thick	standing alone
exact β^*	standing + A5-thick + A8c-attain	standing + exact-contact
robust rat'lity	ditto + TRE-gen-Hall (deterministic)	ditto + menu-Hall (set-valued)
Engine	Balder weak topology on Σ, Mertens minmax, Lusin lift	compactness of $\mathcal{K}(W)$ in finite-dim $\mathbb{R}^{\lvert\Omega\rvert}$

The architectural shift is moving from infinite-dimensional strategy compactness to finite-dimensional payoff-profile compactness. Three substantive consequences:

1. Tier 1a is now *unconditional*. A5-thick (a Lusin-thickness clause that fails in perfect-revelation examples) and A8c-attain (a rowwise-argmin attainment clause that fails on upward-spike rows) are both eliminated for value optimality and ε-adversaries.

2. Exact-contact is endogenous to the optimal labeling, so it can be checked from the chosen w^* rather than from a representative of an abstract maximizer. The sufficient routes (closed

image, closed-graph correspondence, u.h.c. Bayes-action) are all instantiable by primitive choices.

3. Menu-Hall is set-valued, and the support-function form makes the calibration condition a transparent integrated separation inequality. This is strictly weaker than deterministic TRE-gen-Hall (every deterministic m^* kernel is a degenerate κ).

The ternary witness in §8 shows menu-Hall cannot be dropped without further structural conditions on T, but the witness also clarifies the geometric obstruction: it is multi-dimensional vector balance, not stochastic-vs-deterministic. This sharpens what needs to change to push past Tier 2.

10 Remaining directions

1. **When does exact-contact hold automatically?** The sufficient routes listed in §6 cover most natural specifications, but a constructive procedure that always builds a closed-image labeling out of an arbitrary optimal w^* would close Tier 1b unconditionally.

2. **Structural conditions for menu-Hall.** The largest class of trust regions for which menu-Hall holds without further conditions is open. Candidates: separable trust regions, zonotopal T, trust regions invariant under a transitive group action.

3. **Formalization.** A faithful formalization would now encode the menu engine: compactness of $\mathcal{K}(W)$, the menu-value functional, closure-pruning, exact-contact selection, and menu-Hall as a set-valued calibration. This is a substantially lighter formalization target than the Balder/Mertens/Lusin route.

References

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[2] Charalambos D. Aliprantis and Kim C. Border. *Infinite Dimensional Analysis: A Hitchhiker's Guide.* Springer, 3rd edition, 2006. (Specifically: Theorems 18.13 (Kuratowski–Ryll-Nardzewski / Jankov–von Neumann), 18.19 (measurable maximum), 15.11 (compactness of Δ on compact metric).)

[3] Vladimir I. Bogachev. *Measure Theory.* Springer, 2007. (Standard-Borel disintegration; Polish-valued measurable selection.)